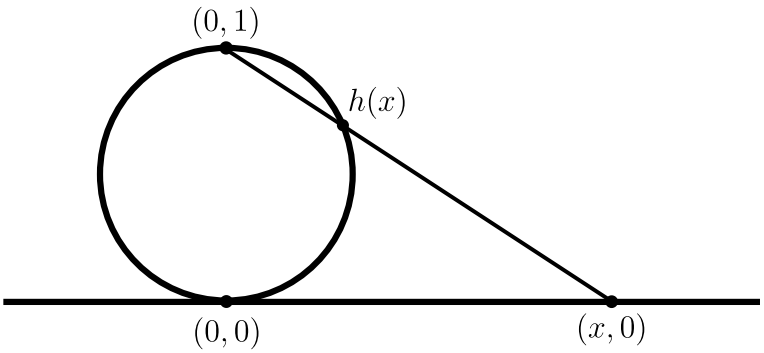
All solutions to graded problems must be typed in LaTeX. All other problems can be typed or written, but must be completed. Graded problems must be organized in numerical order. All problem sets (both graded and non-graded) must be uploaded to Canvas as a single PDF document. Students can find the grading scale in our Syllabus.

Graded

Exercise 1. Let X, d be a metric space with metric d and let $A \subseteq X$. Prove that $d|_{A\times A}$ is metric for the subspace topology on A.

Exercise 2. Let S be the subset of $\mathbb{R}^2$ obtained from the circle centered at $(0, \frac{1}{2})$ with radius $\frac{1}{2}$: $S = \{(x, y) \in \mathbb{R}^2 : x^2 + (y - 1/2)^2 = 1/4\}$. Let X be the subspace of $\mathbb{R}^2$ defined by $X = S - \{(0, 1)\}$. Define $h : \mathbb{R} \to X$ to be the function such that $h(x)$ is the intersection of X with the line through the points $(0, 1)$ and $(x, 0)$ shown below.



1. Let d be the euclidean metric on $\mathbb{R}^2$ and define $\rho : \mathbb{R} \times \mathbb{R} \to \mathbb{R}$ by

$$\rho(x, y) = d(h(x), h(y)).$$

   Prove that ρ is a metric on $\mathbb{R}$ and that the topology induced by ρ is the same as the usual topology on $\mathbb{R}$.

2. Prove that $h : \mathbb{R} \to X$ is a homeomorphism.

Non-Graded

Exercise 3. Let Let X, d be a metric space and let $B_d(x, \varepsilon)$ be the open ball in X of radius $\varepsilon > 0$:

$$B_d(x, \varepsilon) = \{y \in X : d(x, y) < \varepsilon\}.$$

Define the closed ball of radius $\varepsilon > 0$ by

$$\overline{B}_d(x, \varepsilon) = \{y \in X : d(x, y) \leq \varepsilon\}.$$

1. Prove that $\overline{B}_d(x, \varepsilon)$ is closed in X for all x and all $\varepsilon > 0$.
2. Find a counterexample to the statement that $\overline{B_d(x, \varepsilon)} = \overline{B}_d(x, \varepsilon)$ for all metric spaces X, d and all points x, ε. (Reminder: $\overline{B_d(x, \varepsilon)}$ denotes the closure of the open ball $B_d(x, \varepsilon)$ in X)

Exercise 4. Let X be the set of continuous functions $f : [a, b] \to \mathbb{R}$ and define $d^* : X \times X \to \mathbb{R}$ by

$$d^*(f, g) = \int_a^b |f(t) - g(t)| dt.$$

1. Assuming calculus level knowledge of integration, prove that d^* is a metric on X.
2. Define $I : X \to \mathbb{R}$ by
$$I(f) = \int_a^b f(t)\, dt.$$
Prove that when X is considered as the metric space with metric d^* and $\mathbb{R}$ has the standard metric, then I is a continuous function.